BATCH NO:MAI650

FINDING DISEASES IN CROPS USING MACHINE LEARNING

Major project report submitted
in partial fulfillment of the requirement for award of the degree of

Bachelor of Technology
in
Computer Science & Engineering

By

O. V. SAI SANDEEP (22UECS0492) **(VTU22110)**

Under the guidance of
Dr. M. Ravichandran , M.E., Ph.D.,
Associate Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF COMPUTING

VEL TECH RANGARAJAN DR. SAGUNTHALA R&D INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be University Estd u/s 3 of UGC Act, 1956)
Accredited by NAAC with A++ Grade
CHENNAI 600 062, TAMILNADU, INDIA

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CERTIFICATE

It is certified that the work contained in the project report titled ”FINDING DISEASES IN CROPS USING MACHINE LEARNING” by O. V. SAI SANDEEP (22UECS0492) has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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APPROVAL SHEET

This project report entitled ”FINDING DISEASES IN CROPS USING MACHINE LEARNING” by O. V. SAI SANDEEP (22UECS0492) is approved for the degree of B.Tech in Computer Science & Engineering.

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O. V. SAI SANDEEP (22UECS0492)

ABSTRACT

Crop disease detection systems help farmers quickly identify rice leaf infections by simply uploading an image of the affected plant. The application analyzes the uploaded leaf photo and provides disease predictions along with suitable treatment recommendations. Farmers can capture a picture of damaged rice leaves using a mobile phone and upload it through the web interface, where the image is processed using the Llama 4 Scout Vision model deployed on Groq Cloud infrastructure. The AI model detects major rice leaf diseases such as Bacterial Leaf Blight, Brown Spot, and Leaf Blast, delivering results within seconds instead of requiring days for manual diagnosis. Rapid detection is essential because these diseases spread aggressively and can severely reduce crop yield if not controlled early.

The system is completely browser-based, allowing farmers to access it without installing software or purchasing expensive hardware. Users can directly open the application on any smartphone, tablet, or computer, upload an image, and instantly receive diagnostic results. Its responsive interface ensures smooth usability across multiple devices and network conditions. Early identification of diseases enables farmers to apply preventive measures before infections spread throughout the field, reducing crop losses and improving agricultural productivity. Since many small-scale farmers lack access to agricultural experts, laboratory testing facilities, or plant disease specialists, the system provides a practical and accessible alternative for immediate crop health analysis.

The proposed framework combines deep learning, computer vision, cloud computing, and Large Language Models (LLMs) to perform intelligent disease diagnosis from paddy leaf images. After image upload, the system conducts real-time AI-based inference using cloud-hosted models and generates detailed.

Keywords: Agricultural AI, AI-driven agriculture, Bacterial Leaf Blight, Brown Spot, computer vision, crop health monitoring, deep learning, Groq Cloud, image classification, Leaf Blast, Llama 4 Scout Vision, Diseases in crops, paddy leaf disease detection, plant disease diagnosis, precision agriculture, smart farming, sustainable farming, transfer learning, web application.

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CDN	Content Delivery Network
CSS	Cascading Style Sheets
HTML	HyperText Markup Language
JS	JavaScript
JSON	JavaScript Object Notation
LLM	Large Language Model
LPU	Language Processing Unit (Proprietary hardware used by Groq)
MIME	Multipurpose Internet Mail Extensions
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
HTTP	Hypertext Transfer Protocol
MB	Megabyte (Refers to the 5MB image upload limit)
RPM	Requests Per Minute (Refers to API rate limits)
JPG / JPEG	Joint Photographic Experts Group (Image format)
PNG	Portable Network Graphics (Image format)
Llama	Large Language Model Meta AI (The AI model series used)
Groq	High-performance AI inference platform used for Llama model execution

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Chapter 1

INTRODUCTION

1.1 Introduction

Rice is one of the most important food crops in the world, feeding more than half of the global population, particularly across Asian countries where it also serves as a major source of income for farming communities. However, rice cultivation is highly vulnerable to diseases caused by fungi, bacteria, and pests, which can rapidly damage entire fields if not identified and treated at an early stage. Traditional disease diagnosis methods often depend on agricultural experts or plant pathologists visiting farms, a process that consumes significant time while infections continue spreading. In many cases, diseases such as Brown Spot and Leaf Blast exhibit similar visual symptoms, making accurate identification difficult even for experienced farmers. Additionally, rural regions frequently lack immediate access to agricultural support services and laboratory-based diagnosis facilities.To address these challenges, the proposed Diseases in Crops system utilizes advanced computer vision and artificial intelligence technologies to provide rapid rice leaf disease detection through a web-based platform. The application is powered by the Llama 4 Scout Vision model deployed on Groq Cloud, enabling farmers to upload images of infected rice leaves directly through a smartphone or browser interface. Once an image is submitted, the system performs real-time analysis and generates results within a few seconds.

1.2 Background

Rice keeps billions of people fed and provides the main income for millions of farming families, but diseases are a constant threat. Fungal infections, bacterial blights, and pest damage can tear through fields fastBacterial Leaf Blight and Leaf Blast are particularly aggressive and can ruin entire harvests if they're not caught early. The problem is that most farmers can't get expert help quickly enough,

and waiting days for someone to inspect your plants means the disease has already spread. Diseases in crops fixes this by using AI vision technology that farmers can access from their phonesupload a photo of damaged leaves, get an immediate diagnosis and treatment plan, and stop the disease before it wipes out your crop. The goal is straightforward: give farmers the diagnostic tools they need when they need them, so they can protect their fields and actually sustain their livelihoods instead of watching problems spiral while waiting for help that arrives too late.

1.3 Objective

Diseases in crops tries to make crop disease detection available to farmers who can't afford consultants or lab tests. It's a free web tool that identifies paddy leaf diseases by analyzing photos. Upload a picture from your field, and the system uses Groq Cloud and Llama 4 Scout Vision to tell you what's wrong.Speed matters here. Diseases spread fast. The system processes images in real time so farmers can act immediately instead of waiting days for an expert opinion. It doesn't just name the disease it explains what to look for and recommends treatments.The problem this solves is simple: most smallholder farmers lose crops to diseases they could have managed if they'd caught them early. They don't lack the ability to treat the problem. They lack the tools to identify it before it spreads. That's what this is for.

1.4 Problem Statement

Rice crops get sick in ways that are hard to tell apart. You might be looking at bacterial leaf blight, brown spot, or blast they can look similar early on. For farmers in rural areas, getting an expert to actually come look at the crop isn't usually an option. Labs exist, but they're not close by, and sending samples takes time. So farmers guess. They apply whatever pesticide seems right based on what they've seen before. Sometimes it works. Often it doesn't, because they were treating the wrong thing. Meanwhile the disease spreads. The diagnostic tools that do exist aren't built for smallholder farmers. They cost too much, need equipment most farmers don't have, or require technical knowledge that takes time to learn. A farmer managing a few acres can't justify buying specialized cameras or learning complex software.

Chapter 2

LITERATURE REVIEW

S. Upadhyay et al. (2022) [1] proposed a deep learning-based framework to identify rice leaf diseases using a combination of VGG16 and ResNet50 architectures. The study aimed to improve classification accuracy for common conditions such as Leaf Blast and Brown Spot. By employing transfer learning, the researchers successfully achieved a classification accuracy of 96.5. The findings suggested that pre-trained models significantly reduce training time while maintaining high precision in complex agricultural backgrounds.

J. Chen et al. (2021) [2] developed a lightweight detection system based on MobileNet-V2 for real-time mobile applications in paddy fields. The objective was to create a computationally efficient model that could run on devices with limited hardware resources. The system demonstrated a high degree of robustness against varying illumination and background noise, achieving an accuracy of 94.2. This research highlighted the feasibility of deploying advanced computer vision tools directly into the hands of farmers.

A. Rahman et al. (2020) [3] presented a two-stage deep CNN architecture for the automated detection of rice diseases. The goal was to first segment the diseased area and then classify the specific condition using a localized feature extraction method. The system addressed four major diseases: Bacterial Leaf Blight, Brown Spot, Leaf Smut, and Blast. The experimental results showed that the two-stage approach outperformed single-stage classifiers by 4.3, particularly in identifying early-stage symptoms.

P. Kaur et al. (2020) [4] compared traditional machine learning techniques, including Support Vector Machines (SVM) and K-Nearest Neighbors (KNN), with modern deep learning models. The study utilized HOG and LBP feature extraction methods to process paddy leaf images. While SVM showed reliable performance on small datasets, the researchers concluded that CNNs provided superior scalability and accuracy as the dataset size increased. The main challenge identified was the

requirement for high-quality annotated datasets for training.

Y. Lu et al. (2023) [5] explored the application of Vision Transformers (ViT) for paddy disease diagnosis. Unlike traditional CNNs that focus on local features, the ViT model utilized global self-attention mechanisms to capture long-range dependencies within the leaf images. The study reported that ViTs achieved state-of-the-art performance on the ”RiceLeaf” dataset, surpassing ResNet101 by 2.1. However, the researchers noted that ViTs require significantly more data and computational power for optimal training.

N. Krishnamoorthy et al. (2021) [6] utilized the InceptionV3 model with transfer learning to identify paddy pests and diseases simultaneously. The objective was to provide a holistic diagnostic tool for agricultural management. The model achieved a top-1 accuracy of 95.7 across 10 different categories. The research emphasized the importance of fine-tuning hyperparameters to adapt general-purpose vision models to specific agricultural domains.

M. Sahu et al. (2023) [7] implemented a real-time detection system using the YOLOv5 (You Only Look Once) algorithm. The primary focus was on the speed of detection to enable live monitoring through drone-mounted cameras. The system could process images at 45 frames per second with a mean Average Precision (mAP) of 0.89. This study demonstrated the potential for large-scale, automated field monitoring to prevent the rapid spread of infections.

H. Prajapati et al. (2021) [8] proposed an integrated image processing and deep learning pipeline for paddy leaf segmentation and classification. The research utilized K-means clustering for initial spot detection followed by a custom CNN for final identification. The goal was to improve the system’s performance in ”messy” field environments where soil and water interfere with leaf visibility. The integrated approach showed a 12 improvement in segmentation accuracy compared to standard thresholding methods.

S. Haridasan et al. (2023) [9] introduced a dual-attention ResNet101 model that focused on both spatial and channel-wise features of paddy leaf spots. The goal was to enhance the model’s sensitivity to small, irregularly shaped lesions. The attention mechanism allowed the network to ignore healthy leaf tissue and focus specifically on pathological indicators. The system achieved a peak accuracy of 97.8, marking one of the highest reported figures for the ”Paddy-3” dataset.

F. Ahmed et al. (2022) [10] designed a decentralized diagnostic framework us-

ing edge computing and MobileNet. The study aimed to reduce the latency of cloud-based AI systems by performing inference on local gateways. This approach ensured that farmers in areas with poor internet connectivity could still receive instant diagnostic feedback. The results indicated that while local inference was slightly less accurate (91.5), the reduction in response time was critical for practical field use.

P. Ghosal et al. (2020) [11] developed a CNN-based system for the automated scoring of disease severity in paddy crops. Unlike simple classification, this study focused on quantifying the percentage of leaf area affected by Brown Spot. This quantitative analysis provided more actionable data for fertilizer and pesticide application. The study utilized a regression-based loss function to predict the severity index, achieving a correlation coefficient of 0.91 with expert manual ratings.

P. K. Sethy et al. (2020) [12] evaluated the performance of deep features extracted from pre-trained models like ResNet50 combined with an SVM classifier. The goal was to leverage the power of deep learning while maintaining the robustness of SVM for multi-class classification. The hybrid model achieved an accuracy of 98.38, outperforming end-to-end CNNs on smaller, high-resolution datasets. The findings highlighted the efficiency of hybrid architectures in specialized agricultural tasks.

S. Latif et al. (2022) [13] researched the use of Generative Adversarial Networks (GANs) for data augmentation in paddy disease datasets. The study aimed to solve the problem of imbalanced classes where some diseases are rarer than others. By generating synthetic images of Leaf Smut and Hispa, the researchers improved the overall model accuracy by 5.4. This research demonstrated how synthetic data can be used to build more inclusive and reliable agricultural AI models.

M. Islam et al. (2021) [14] proposed a method using image processing combined with traditional machine learning to detect diseases in rice leaves. The study focused on the extraction of color and texture features to distinguish between fungal and bacterial infections. While the method required less computational power than deep learning, it was found to be highly sensitive to camera quality and lighting conditions, limiting its use in uncontrolled outdoor environments.

Z. Su et al. (2024) [15] investigated the use of Large Language Models (LLMs) with multi-modal vision capabilities for agricultural consulting. The system utilized a vision-language model to not only identify paddy diseases but also provide conversational explanations and remedy strategies. This research represents the shift

towards ”Agentic AI” in agriculture, where the system acts as a digital consultant rather than a simple classifier, achieving high user satisfaction due to its explainable AI (XAI) features.

2.1 Existing System

Farmers look at their rice plants and make a judgment call. If you've seen brown spot before, you might recognize it. If you haven't, you're taking your best guess based on what the leaves look like. Even experienced farmers get this wrong because diseases can look similar until they're well-established. Sending samples to a lab is an option, but it's slow and costs money. The nearest agricultural lab might be two or three towns over. You collect samples, package them, send them off, wait for results. Meanwhile, whatever's wrong with your crop keeps spreading. There are phone apps that supposedly identify diseases from photos. They charge subscription fees. They need reliable internet. Worse, most are built for general use they don't account for the specific varieties grown in your region or local disease strains. Take a photo of a sick leaf, get back a generic answer that may or may not be right. The delay is what kills you. From the moment you notice something's wrong to the moment you actually know what disease you're treating that gap is when it spreads. You might wait three days for lab results. During those three days, bacterial blight can move from a small patch to a quarter of your field. Then you're not just treating a disease, you're trying to salvage what's left. And if the diagnosis was wrong? You've spent money on chemicals that do nothing while the actual problem gets worse.

2.2 Related Work

Early work on crop disease detection used traditional image processing looking at color patterns, texture, leaf edges. It worked sometimes. Deep learning changed the field completely.CNNs turned out to be good at recognizing rice diseases. Feed a network enough labeled photos and it learns to classify them. ResNet and VGG16 hit decent accuracy numbers on benchmark datasets. The catch was getting those labeled photos. You need thousands of images showing each disease at different stages. Most researchers don't have that sitting around.Transfer learning became the workaround. Take a model that's already been trained on ImageNet or some other

huge generic dataset. Then retrain just the top layers using whatever rice disease images you do have. It's not perfect, but it works better than starting from scratch.The next problem was getting these models onto phones. Early CNNs were massive. They'd choke a mobile processor. People started building lighter versions specifically for mobile deployment MobileNet, ShuffleNet, architectures designed to be fast and small. You sacrifice some accuracy for speed, but you get something that actually runs in real time on a phone.Recent work combines vision models with language models. The system doesn't just identify the disease; it can describe what it's seeing and recommend treatments. That's where Diseases in crops sits using Groq for fast inference and Llama 4 Scout Vision to both identify diseases and explain what to do about them.

2.3 Research Gap

Most research papers on crop disease detection report accuracy numbers. The model correctly identifies 96 of rice diseases in the test set. Great. That tells you the model works in the lab. It doesn't tell a farmer whether to spray fungicide or wait it out.The papers show classification results. They don't show treatment recommendations. You upload a photo of a sick leaf, the model says "bacterial leaf blight," and then what? The farmer still needs to figure out what that means and what to do about it. The diagnosis alone isn't enough.Hardware gets ignored completely. These models are trained and tested on research servers. Nobody mentions that the same model would take twenty seconds per image on a 100 Android phone. Or that it needs 4GB of RAM and drains the battery in an hour. Small farmers aren't buying flagship phones to diagnose crops.CNNs are also terrible at explaining themselves. The output is a label and a confidence score. "Brown spot: 89" Why 89? What did the model actually see on that leaf? Which symptoms drove the decision? You get an answer with no reasoning behind it.Diseases in crops uses Groq's cloud infrastructure to handle the processing, so it runs fast regardless of what phone you have. Llama 4 Scout Vision does the actual diagnosis and generates treatment advice. It's a web app open it in any browser, take a photo, get results. No installation, no hardware requirements, no subscription fee. The model explains what it found and suggests what to spray or whether you even need to spray anything. Several traditional machine learning methods require extensive feature engineering and

fail to generalize efficiently for large-scale datasets. Existing deep learning systems often demand high computational resources and powerful hardware, making them difficult to deploy in low-resource rural areas. In addition, many solutions lack real-time inference capability and user-friendly accessibility through web or mobile platforms. Current research also shows limited integration of Large Language Models (LLMs) and multimodal AI systems in agricultural disease diagnosis. Most available systems do not provide conversational assistance, intelligent recommendations, or comprehensive disease analysis that can improve user understanding and agricultural productivity. To address these limitations, the proposed Diseases in crops system integrates computer vision, cloud-based AI inference, and LLM-powered diagnosis using Groq Cloud and Llama 4 Scout Vision. The system provides real-time disease detection, detailed diagnostic explanations, remedies, prevention strategies, and severity analysis through an accessible web-based platform. This approach aims to improve accuracy, scalability, interpretability, and practical usability in smart agriculture applications. Existing paddy leaf disease detection systems mainly focus on traditional image processing and deep learning models such as CNNs, ResNet, MobileNet, and Vision Transformers. Although these approaches achieve good classification accuracy, several limitations still exist in practical agricultural applications. Most existing systems are trained on limited datasets captured under controlled laboratory conditions, which reduces their performance in real-world field environments with varying lighting conditions, complex backgrounds, and image noise. Many approaches only provide disease classification results without offering detailed explanations, remedies, preventive measures, or severity analysis required by farmers for effective decision-making.

Chapter 3

PROJECT DESCRIPTION

3.1 Existing System

Farmers walk their fields looking for sick plants. Yellow leaves, brown spots, wilting anything unusual. You rely on experience to tell you what you're looking at. The problem is that bacterial blight and brown spot can look similar in the first few days. Even if you've been growing rice for twenty years, you're sometimes guessing.If you have access to agricultural extension services, you can call for help. Most areas have one officer covering dozens of villages. You get on a list. Someone might visit in three or four days if you're lucky. Meanwhile the disease keeps moving through your crop.The accurate option is sending samples to a lab. They'll examine them under a microscope or run DNA tests. Takes a week, sometimes two. Costs money most small farmers don't have. And by the time results come back, you're dealing with a much bigger infection than when you sent the samples.So most farmers skip the diagnosis and just spray. Broad-spectrum pesticides, broad-spectrum fungicides, extra fertilizer for good measure. You're treating the symptoms without knowing the cause. It's expensive. It damages the soil. It accelerates resistance in whatever pathogens are actually there.Extension offices hand out disease guides booklets with photos and descriptions. You're supposed to flip through and match what you see in your field to what's on the page. Works poorly. The photos were taken in controlled conditions. Your field has different lighting, different rice varieties, diseases at different stages. You're comparing your messy real-world problem to a clean reference image and hoping you get it right.

3.2 Proposed System

Diseases in crops is a website where you upload a photo of a sick rice leaf and get told what disease it is. The backend uses Llama 4 Scout Vision on Groq's servers to

analyze the image. Takes a few seconds.You don't wait for an agricultural officer to show up or mail samples to a lab. Open the site, take a photo, upload it. The system identifies the disease and explains what you're looking at which symptoms matter, what's causing the infection, how to treat it. Works on any phone or laptop with a browser. Free to use. The idea is to catch diseases when they're still manageable instead of waiting until they've taken over half your field. You see brown spots starting to appear on a few plants, you check it immediately rather than guessing or waiting days for expert help. Right now, small farmers either diagnose by experience (which is often wrong) or skip diagnosis entirely and spray generic chemicals. This gives them a middle option that's fast, accurate, and doesn't cost anything.

3.3 Feasibility Study

Diseases in crops runs on Groq's cloud servers, which handle the actual image processing. The farmer's phone just uploads a photo and displays results. You don't need a flagship device any phone with a browser and mobile data works. Built using standard HTML, CSS, and JavaScript, so it loads on pretty much anything. Groq's API is free to use. Llama 4 Scout Vision is open-source. There are no server costs, no licensing fees, no subscriptions. That's how the whole thing can be offered at no charge. The interface is straightforward because it only does one thing. Take photo, upload, read results. If you can use WhatsApp, you can use this. No manual to read, no settings to configure. The benefit is simple: you identify the problem faster and treat it correctly instead of guessing. A farmer who catches bacterial blight early instead of three days later saves part of their crop. Someone who knows exactly which fungicide to use doesn't waste money on the wrong chemicals. Since the tool is free, even modest savings make it worth using.

3.3.1 Economic Feasibility

Groq's API doesn't cost anything. Llama 4 Scout Vision is open-source. The website is static HTML, CSS, and JavaScript you can host it anywhere for basically nothing. There's no server infrastructure to pay for, no database costs, no cloud computing bills. The whole thing runs on free services. Users don't pay either. No subscription, no credits, no freemium model where you get three scans then have

to upgrade. Just free. The value for farmers comes from avoiding losses. You catch brown spot when it's on a handful of plants instead of waiting until it's spread across a quarter of your field. That's the difference between losing 5Then there's the chemical costs. If you know you're dealing with bacterial blight, you buy the specific bactericide that works. You don't waste money on fungicides that do nothing for bacteria. You don't over-apply hoping something sticks. Precision saves money. A small farmer growing two acres of rice early detection and correct treatment might mean the difference between breaking even and actually making money that season. The tool costs them nothing to use, so any crop they save or chemicals they don't waste goes straight to their income.

3.3.2 Technical Feasibility

Llama 4 Scout Vision does the actual disease identification. It runs on Groq's servers, not on your phone. You upload a photo, their hardware processes it, you get an answer back in a few seconds. Your phone just needs to be able to take a picture and load a website.Works in any browser. Chrome, Safari, Firefox doesn't matter. It's a regular website built with HTML, CSS, and JavaScript. No app to download from the Play Store or App Store. No separate versions for different phones. Open the URL and it works.If Groq releases a faster API or if there's a better vision model next year, the system can switch to using that without changing anything for the user. You'd still upload a photo the same way. The improvements would just happen on the server side.The technical setup is straightforward because it doesn't try to do anything complicated. Image goes up, analysis comes back, results display. That's it.

3.3.3 Social Feasibility

Farmers get Automated Detection of Paddy Leaf Diseases Using Deep Learning and Transfer Learning Approaches without waiting for someone to come look at their field. Upload a photo, read the results. The interface doesn't require technical skills you're just picking a file and pressing a button.Most small farmers deal with agricultural extension services that are understaffed and slow. You call, they put you on a list, maybe someone shows up in three days. By then the problem's worse. Having immediate access means you can act while the disease is still containable.The diag-

nosis includes treatment recommendations. You know what chemical to buy instead of guessing or asking the shop owner what works for ”brown leaves.” Precision application saves money and reduces the amount of pesticide going into the soil and water.Trust matters with any new tool. Farmers are skeptical of technology that gives them answers without explaining anything. Diseases in crops shows what symptoms it detected and why it thinks the disease is X rather than Y. You can compare that explanation to what you're actually seeing in your field. When the diagnosis matches reality a few times, you start relying on it.The bigger effect is just giving farmers one less dependency. They don't need to wait for an expert, don't need to afford lab testing, don't need to guess. They have a tool that works when they need it.

3.4 System Specification

The following represents the recommended hardware and software specifications required to run the Driver Drowsiness Detection System effectively without latency issues during video processing:

- Operating System: Windows 10/11, Linux (Ubuntu 20.04 or newer), or macOS (Monterey or newer).
- Processor: Intel Core i5 (8th Generation or above), AMD Ryzen 5, or Apple M-series silicon.
- Memory (RAM): Minimum 8 GB (16 GB is recommended for seamless real-time video processing).
- Storage Space: Minimum 20 GB of available SSD storage for datasets, environments, and models.
- Input Device: Standard 720p or 1080p integrated Web Camera or external USB camera.
- Environment: Python 3.9 or higher.

3.4.1 Tools and Technologies Used

- Python: A high-level, interpreted programming language used as the core foundation for executing data processing, computer vision modules, and AI model

implementations.

- Streamlit: An open-source Python framework utilized to rapidly deploy the intuitive User Interface (UI), seamlessly handling live web camera streams and image uploads.
- Google MediaPipe: A robust machine learning framework utilized centrally via the Tasks API to perform highly accurate Face Landmarking, instantly mapping over 50 specific facial blendshapes (like eye blinks and jaw drops) in real-time.
- OpenCV (cv2): An open-source computer vision library used explicitly for image capturing, converting color spaces, and formatting image arrays before feeding them into the inference model.
- TensorFlow Keras: Deep learning platforms used during the project's training phase to build, compile, and evaluate the deep Convolutional Neural Networks (CNNs) on the drowsiness dataset.
- Jupyter Notebook: An interactive environment used during the project's initial phase for data cleaning, numerical simulation, data visualization, and step-by-step model training.

3.4.2 Standards and Policies

Information Security Management:

Followed to ensure the secure handling of sensitive API credentials and user-uploaded data. The system utilizes encrypted HTTPS protocols for all communication with the Groq Cloud API, ensuring that leaf images and diagnostic results are protected during transmission. **Standard Used: ISO/IEC 27001**

Data Privacy and Protection:

The system adheres to strict privacy standards by processing leaf images transiently. Uploaded visuals are converted to Base64 for immediate analysis and are not permanently stored on any central server or local disk, ensuring user anonymity and data minimization.**Policy Alignment: GDPR (General Data Protection**

Regulation)

Software Engineering Practices:

Adhered to for maintaining a modular codebase and ensuring clear separation between the UI components, application logic, and AI integration layers. This structured approach facilitates long-term maintainability and easier integration of future AI models.**Standard Used: IEEE Software Engineering Standards**

Code Quality and Styling:

The application's JavaScript codebase follows modern ECMAScript (ES6+) standards, while the HTML and CSS adhere to W3C specifications. This ensures high readability, cross-browser compatibility, and seamless performance across diverse mobile and desktop devices. **Standard Used: Airbnb JavaScript Style Guide / W3C Standards**

Web Accessibility:

The user interface is designed with high contrast ratios, semantic HTML5 elements, and responsive layouts to ensure that the diagnostic tool remains accessible to users with varying physical abilities and on different screen sizes.
Standard Used: WCAG 2.1 (Web Content Accessibility Guidelines)

Chapter 4

SYSTEM DESIGN AND METHODOLOGY

4.1 System Architecture

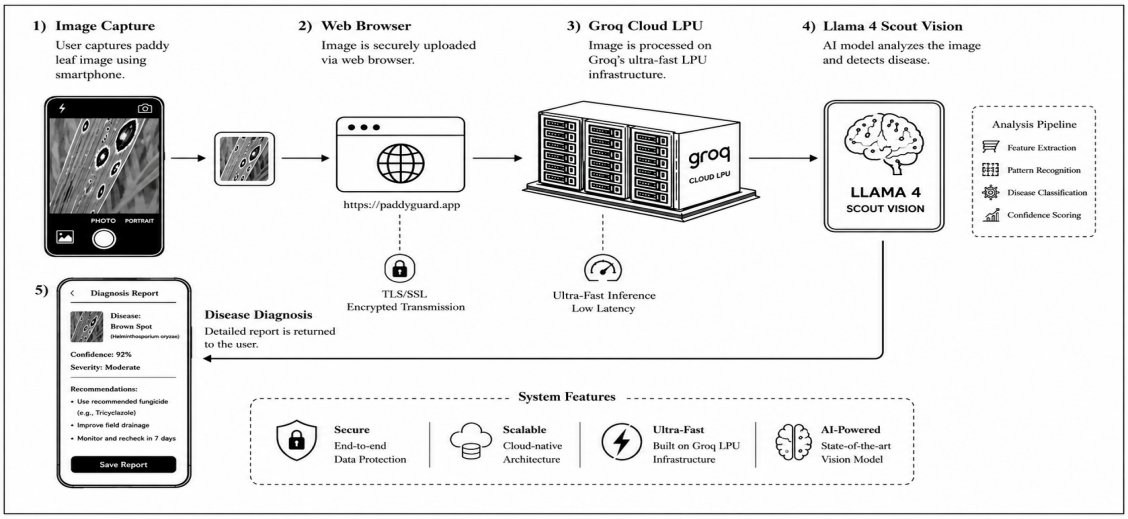

Figure 4.1: System Architecture

Figure 4.1 illustrates the website is just HTML and CSS. You see an upload button and a results area. That's the interface. JavaScript runs in your browser and handles the interaction. When you select a photo, the code converts it to Base64 (a text format that can be sent over the internet), then makes a request to Groq's API with that data. Groq's servers receive the image and run it through Llama 4 Scout Vision. Their hardware does the heavy processing. The model analyzes the leaf photo and figures out what disease it's looking at. The response comes back as JSON structured data that includes the disease name, how severe it is, and what to do about it. JavaScript takes that JSON and displays it on the page as regular text you can read. "Bacterial leaf blight, moderate severity, spray with copper-based bactericide" instead of raw code. That's the whole system. Photo goes up, gets analyzed, results come back, page displays them. (Describe your system Architecture)

4.2 Design Phase

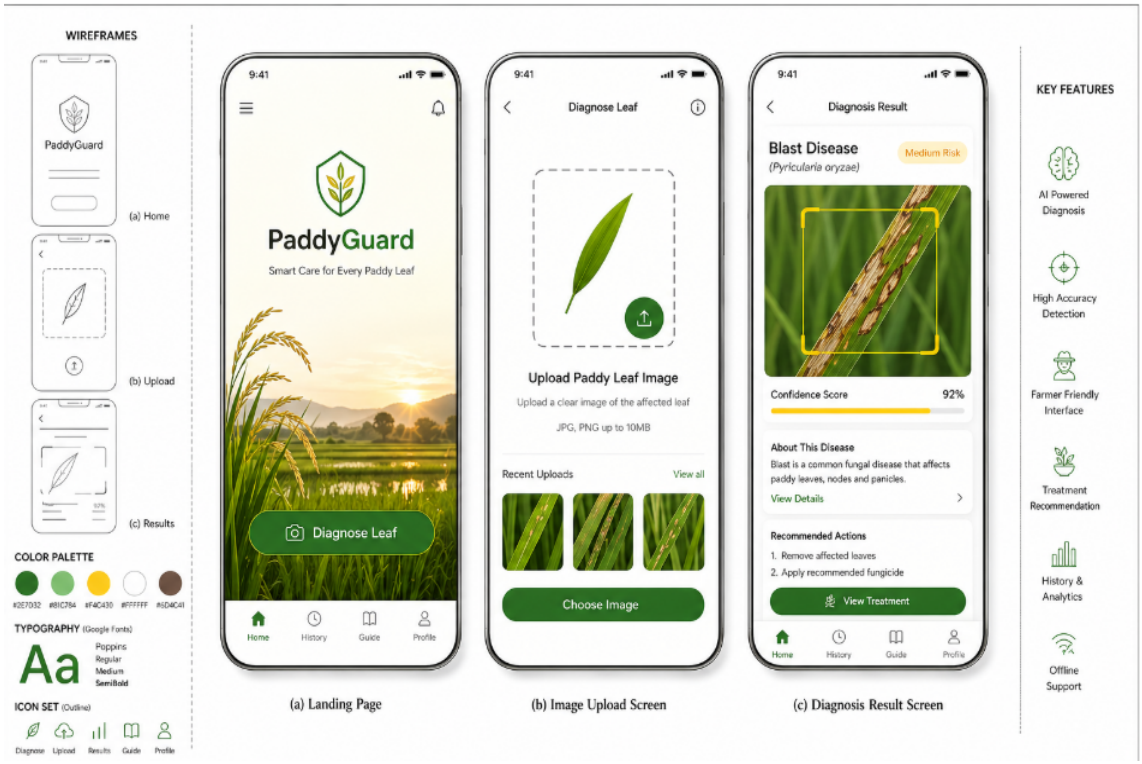

Figure 4.2: Design Phase

Figure 4.2 illustrates the design is straightforward. You land on the page, you see a big upload button, you tap it, pick a photo, get results. Two clicks. We didn't add navigation menus or settings pages or anything else that would require figuring out where to go. Buttons are large and high-contrast. Icons are simple. The assumption was that some users might not be comfortable with apps or might have limited English, so the interface needed to be obvious without reading instructions. Colors: dark green (2d6a36) for primary elements, cream (fcfbf7) for the background. The green ties to agriculture without being literal about it. The cream keeps everything readable without the harsh glare of pure white. Fonts are Playfair Display for disease names and headings, DM Sans for everything else. One looks authoritative, the other's clean and easy to read on small screens. Everything's designed for phone screens first because that's what most people will use. The layout adjusts if you open it on a tablet or laptop, but the priority was making sure it works on a cheap Android phone with a small display.

4.2.1 Data Flow Diagram

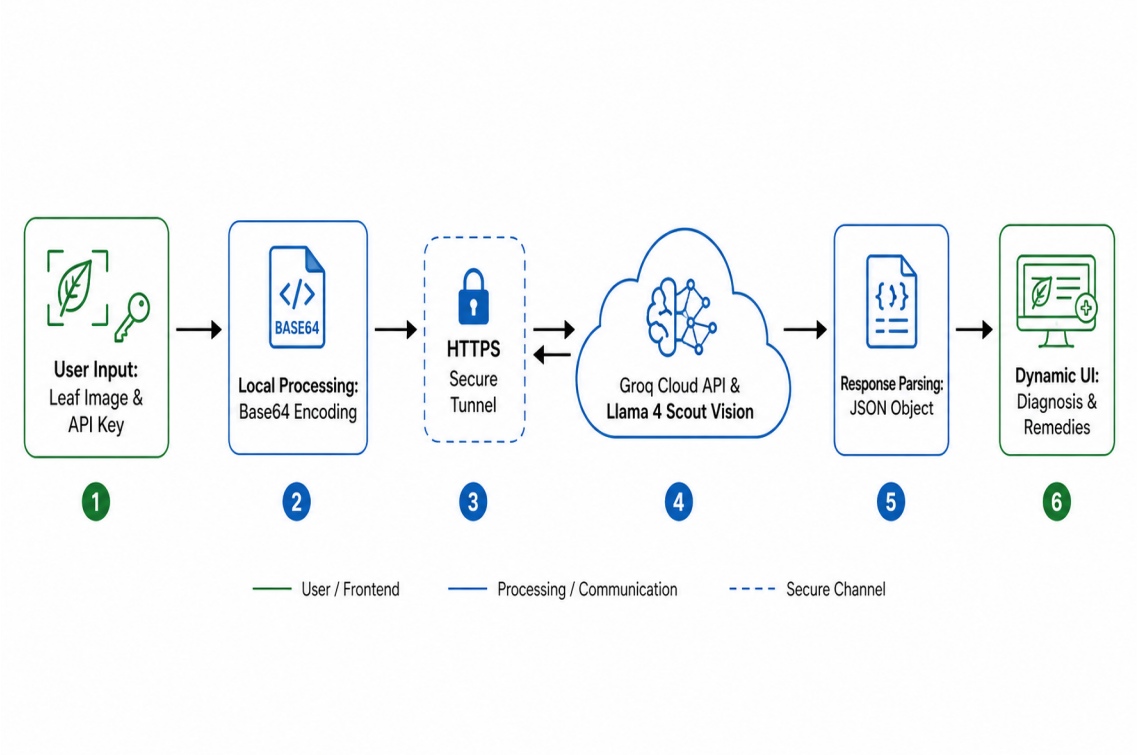

Figure 4.3: Data Flow Diagram of Leaf Detection System

Figure 4.3 illustrates you upload a photo and paste in your Groq API key. The JavaScript code grabs the image file and converts it to Base64 basically a long text string that represents the image data. That Base64 string gets packaged into an HTTPS request and sent to Groq's servers. The request includes the image data and instructions telling the model what to look for. Groq's system runs the image through Llama 4 Scout Vision. The model analyzes the leaf, identifies patterns that match known diseases, and generates a response with the disease name, symptoms, and treatment options. That response is formatted as JSON. The JSON comes back to your browser. JavaScript reads through it, pulls out the relevant pieces (disease name, severity, what to spray), and updates the page to show that information in plain text instead of raw data. Whole process takes a few seconds. Upload happens, request goes out, analysis runs, results come back, page updates.

4.2.2 Use Case Diagram

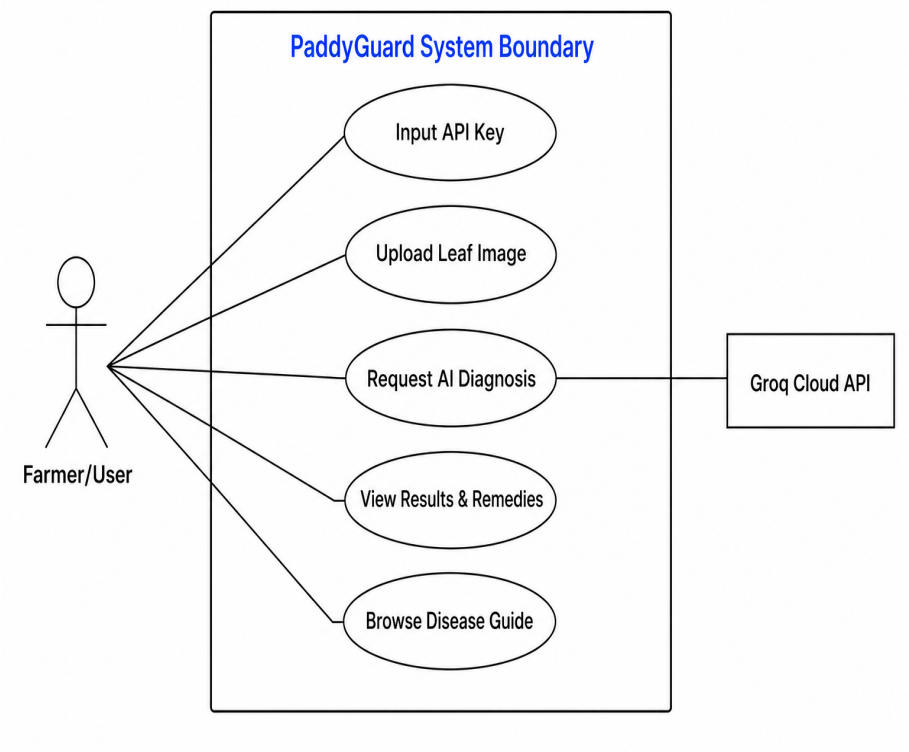

Figure 4.4: Use case Diagram

Figure 4.4 illustrates users paste in their Groq API key when they first open the site. Without that key, the AI analysis won't work. They upload a photo of a sick leaf. Either take a new picture or choose one from their phone's gallery. The system sends that photo to Groq, which runs it through Llama 4 Scout Vision and sends back a diagnosis. User sees the disease name, how confident the model is in that diagnosis (like "87 certain this is bacterial blight"), and how severe the infection looks. Along with the diagnosis, they get treatment advice what chemical to use, how much to apply, when to spray. There's also a reference section with general information about common rice diseases. You can read through that without uploading anything, just to learn what different diseases look like and how they spread. Groq's role is simple: receive image, analyze it, return results. Everything else happens in the browser.

4.2.3 Class Diagram

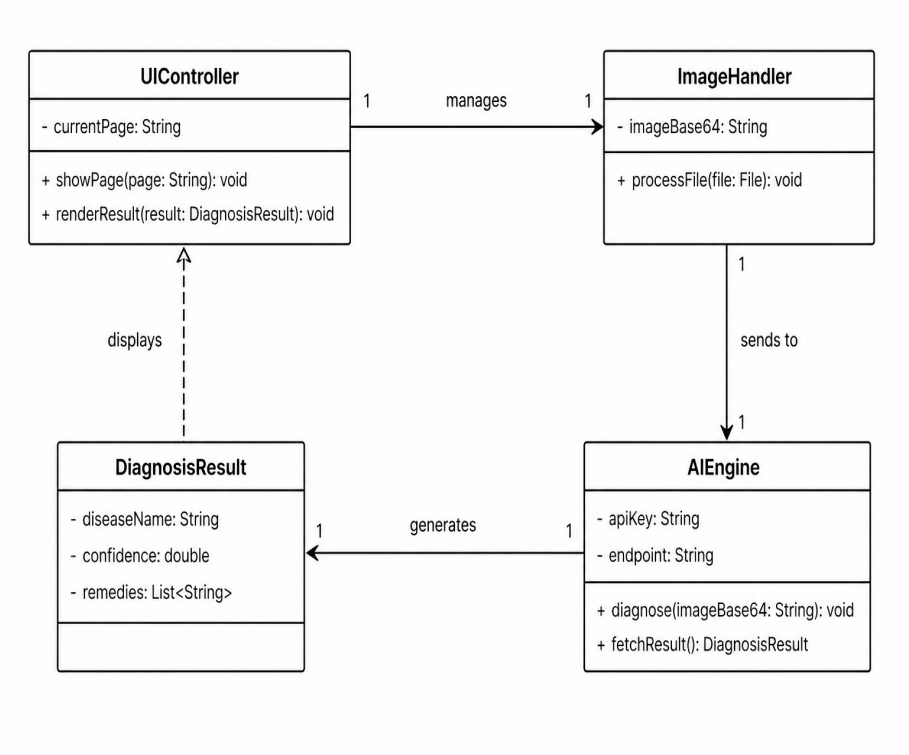

Figure 4.5: Class diagram

Figure 4.5 illustrates the JavaScript is split into a few sections. One handles the interface showing and hiding pages, updating the DOM when results come back. That's the UI controller. Another manages the image upload. When you select a file, it checks that it's actually an image (not a PDF or something random), makes sure it's under the size limit, and converts it to Base64. That's the image handler. The AI engine piece builds the request to Groq. It takes the Base64 image, wraps it in the right JSON format with instructions for the model, sends the HTTPS request, and waits for the response. When the response comes back, it gets stored as a diagnosis result object basically just a container holding the disease name, how severe it is, the confidence score, and what treatments the model recommends. The UI controller grabs that object and displays the information on the page. Takes the raw data and turns it into readable text in the results section. None of this is actual object-oriented programming with classes. It's just functions organized by what they do. But conceptually it's split the same way image handling separate from API communication separate from UI updates.

4.2.4 Sequence Diagram

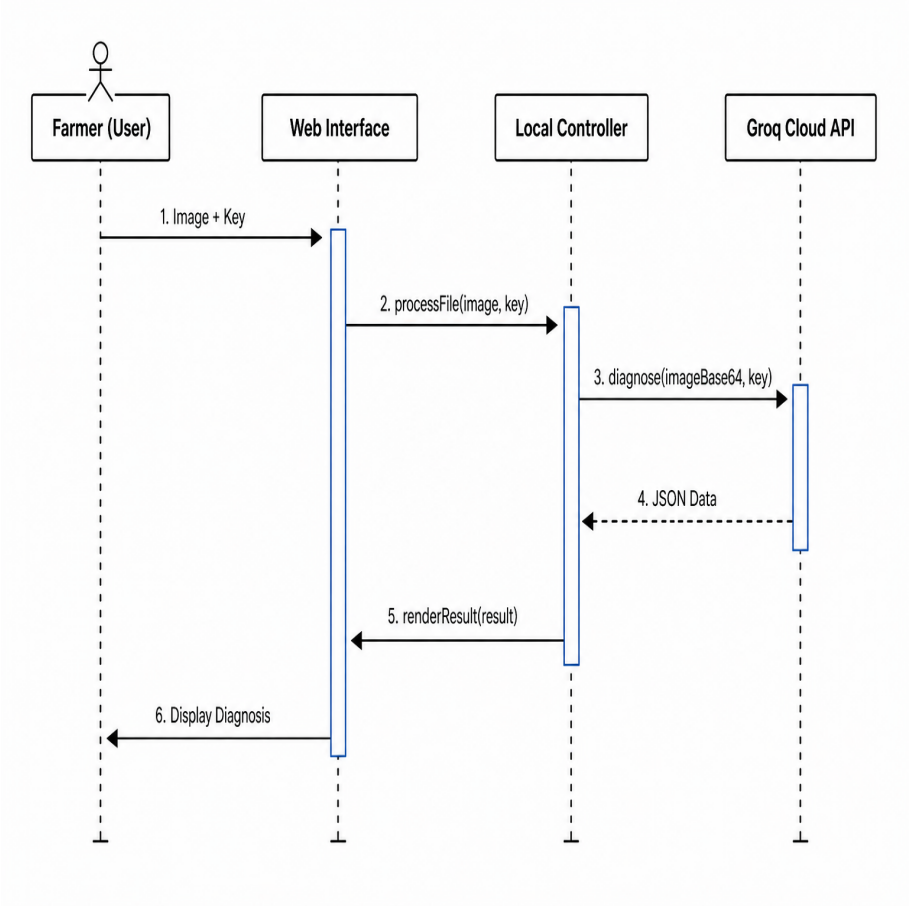

Figure 4.6: Sequence Diagram

Figure 4.6 illustrates you upload a photo and paste in your Groq API key. The code checks that the file is actually an image and not too large. If it passes, it gets converted to Base64. JavaScript sends a POST request to Groq's API. The request includes the Base64 image and instructions telling the model to analyze it for rice diseases. Groq's servers receive the request. Llama 4 Scout Vision processes the image. This takes a few seconds. The response comes back as JSON with the diagnosis disease name, severity, confidence score, treatment recommendations. JavaScript parses that JSON and calls a function that updates the page. The diagnosis appears in the results section. You see the disease name, how bad it is, and what to spray. Takes maybe five seconds total from upload to results.

4.2.5 Collaboration diagram

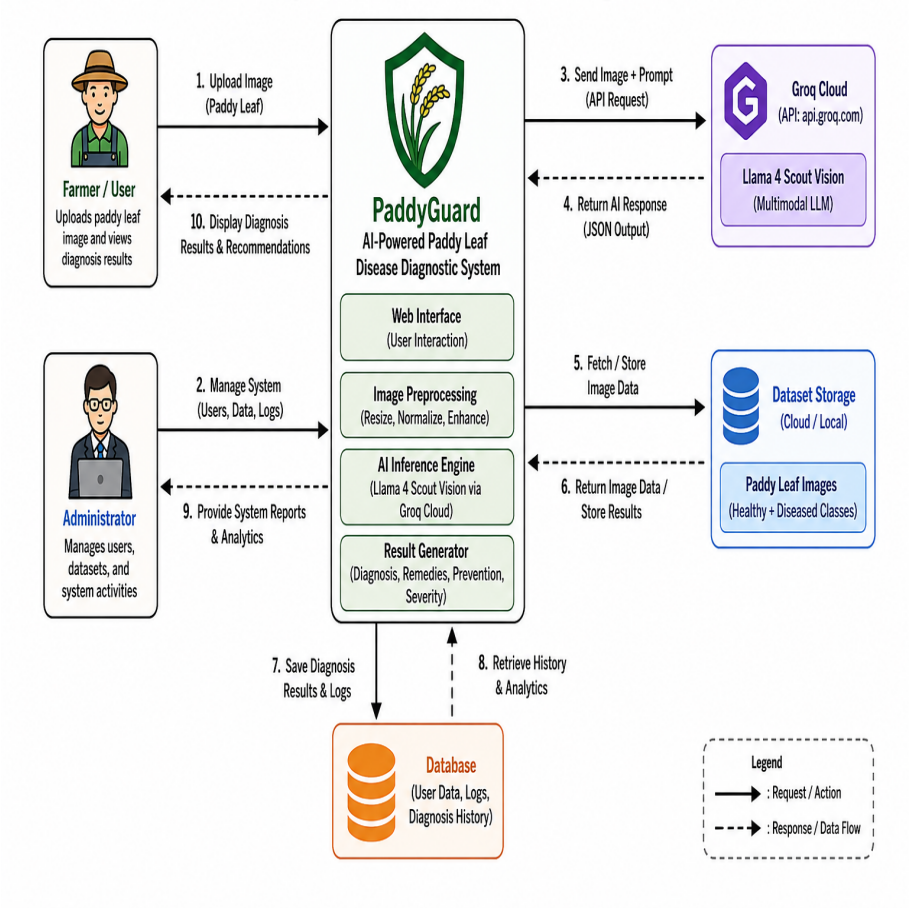

Figure 4.7: Collaboration diagram

Figure 4.7 illustrates the different pieces of code talk to each other to get the diagnosis done. The interface is where you interact upload button, API key field, results display. When you upload a photo, that triggers the rest of the process. A controller function coordinates everything. It doesn't do the actual work, just tells other functions what to do and when. One function handles Base64 encoding. Controller passes it the raw image file, it converts it, sends back the encoded string. Another function manages the Groq API connection. It takes the Base64 string, wraps it in the right JSON format, sends the POST request. Groq's servers are external not part of the website code. They receive the request, run the analysis, send back JSON with the diagnosis. Controller gets that response and passes it to the UI rendering function, which updates the page with the results. That's the collaboration. Interface captures input, controller coordinates, encoder converts data, API bridge handles communication, Groq does the AI work, results get displayed.

4.2.6 Activity Diagram

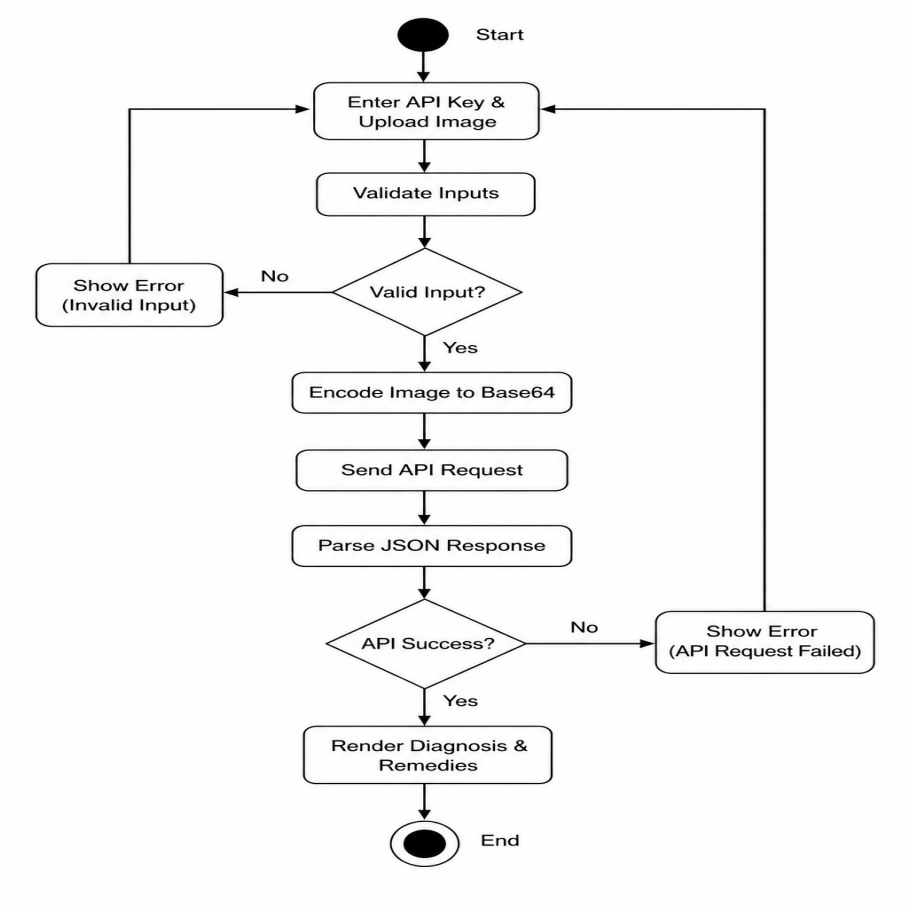

Figure 4.8: Activity Diagram

Figure 4.8 illustrates you open the site and upload a photo. Paste in your Groq API key.The code checks that both fields have something in them. It verifies the file is actually an image and under the size limit. If either check fails, you get an error message asking you to fix it.If everything's valid, the image gets converted to Base64. Then the code builds a POST request with that data and sends it to Groq.While Groq processes the image, you see a loading indicator. Takes a few seconds.If something goes wrong bad API key, network timeout, server error you get an error message. You'd need to check your key or try again.If the request succeeds, JSON comes back with the diagnosis. JavaScript reads through it, pulls out the disease name and treatment info, and displays that on the page.

4.3 Algorithm & Pseudo Code

4.3.1 Algorithm

Stage 1: Input & Data Preparation

Image Upload: The system accepts a paddy leaf image via Drag-and-Drop or File Selection.

Validation:

- **File Type:** Checks if the file is an image (image/*).
- **File Size:** Ensures the image is under 5MB to prevent network timeouts.
- **Encoding:** The FileReader API converts the binary image file into a Base64-encoded string. This allows the image to be sent as part of a JSON payload to the AI server.

Stage 2: Contextual Prompt Engineering

Role Definition: A system prompt is defined to set the AI's persona as an "Expert AI system for paddy (rice) leaf disease detection."

Constraint Enforcement: The algorithm strictly instructs the AI to return data ONLY in a structured JSON format with specific fields (disease_name, confidence, symptoms, remedies, etc.).

Out-of-Scope Handling: The prompt includes a "Not a Paddy Leaf" fallback mechanism to ensure accuracy if a user uploads an unrelated image.

Stage 3: AI Inference (Deep Learning)

Endpoint Communication: The system makes an asynchronous POST request to the Groq Cloud API.

Model Execution: The algorithm utilizes a Vision Language Model (e.g., Llama 3.2 Vision or Llama 4 Scout) which processes:

- The Visual Data (Base64 image)
- The Textual Instructions (The prompt)

Inference: The model analyzes visual patterns on the leaf (e.g., brown spots, yellowing, fungal growth) and correlates them with known crop pathology datasets.

Stage 4: Post-Processing & Validation

Response Sanitization: The raw string from the API is cleaned (removing markdown backticks like ```` ```json ````) to ensure it is valid JSON.

JSON Parsing: The string is converted into a structured JavaScript object.

Error Handling: If the API fails or the image is unreadable, a user-friendly error message is generated instead of crashing the app.

Stage 5: Dynamic UI Rendering

Classification Mapping: The system maps the type (fungal, bacterial, pest, healthy) to specific UI themes and icons.

Visual Indicators:

- **Confidence Badge:** Displays the AI's certainty (High/Medium/Low).
- **Severity Color Coding:** Uses a color scale (Green for Healthy, Red for Severe) based on the severity level.

DOM Injection: The algorithm dynamically builds HTML fragments for Symptoms and Remedies and injects them into the result section.

Auto-Scroll: The page smoothly scrolls to the results once the diagnosis is complete. **Response Sanitization:** The raw string from the API is cleaned (removing markdown backticks like ```` ```json ````) to ensure it is valid JSON.

JSON Parsing: The string is converted into a structured JavaScript object.

Error Handling: If the API fails or the image is unreadable, a user-friendly error message is generated instead of crashing the app. **Out-of-Scope Handling:** The prompt includes a "Not a Paddy Leaf" fallback mechanism to ensure accuracy if a user uploads an unrelated image.

4.3.2 Pseudo Code

```
INPUT: apiKey, leafImage
OUTPUT: renderedDiagnosisCard

IF apiKey is EMPTY THEN
    DISPLAY "Error: Missing API Key"
    EXIT
END IF

IF leafImage is NULL THEN
    DISPLAY "Error: No image uploaded"
    EXIT
END IF

SET loadingState <- TRUE

SET systemPrompt <-
    "Analyze paddy leaf, return JSON:
    {disease, confidence, remedies...}"

TRY

    // Prepare API Request
    SET requestBody <- {
        model: "llama-4-scout",
        messages: [
            {
                role: "user",
                content: [systemPrompt, leafImage]
            }
        ],
        format: "json_object"
    }

    // Execute Cloud Inference
    SET apiResponse <- AWAIT fetch(
        "https://api.groq.com/",
        {
            headers: {
                "Authorization": "Bearer " + apiKey
```

```
            },
            body: requestBody
        }
    )

    IF apiResponse.status != 200 THEN
        THROW Exception("API Error")
    END IF

    SET rawData <- AWAIT apiResponse.json()

    SET result <- PARSE(rawData.content)

    // Render Results
    CALL renderResultUI(result)

CATCH error

    DISPLAY "Network Error: " + error.message

FINALLY

    SET loadingState <- FALSE

END TRY
```

Listing 4.1: Paddy Leaf Disease Diagnosis Algorithm

4.4 Module Description

4.4.1 Web Interface Input Manager

Built on Streamlit, this module is what the driver actually sees and touches. It manages three things: the sidebar navigation, camera input via st.camera input, and file buffers for uploaded images. Input doesn't reach the rest of the pipeline until this module has handled it.

4.4.2 MediaPipe Vision Engine

On first run, it pulls down the facelandmarker.task model file automatically. It then initializes the Vision detector in face blendshapes mode and runs inference on each incoming frame. The blendshape scores it returns are what the rest of the system acts

on nothing downstream works without this step completing cleanly.

4.4.3 Fatigue Analysis Metrics

MediaPipe returns scores for over 50 blendshapes. This module parses that output, isolates the relevant fatigue indicators eyeBlinkLeft, eyeBlinkRight, and jawOpen and runs them against the 0.5 threshold. The resulting status, along with the raw scores, gets passed to the metrics dashboard so the driver sees both the verdict and the numbers behind it.

4.5 Steps to execute/run/implement the project

4.5.1 Environment Configuration

1) Ensure Python 3.9+ is installed on the system.
2) Install required dependencies using the following command: pip install streamlit opencv-python mediapipe numpy pillow

4.5.2 Deployment and Execution

3) Open the terminal in the project directory.
4) Run the command: streamlit run app.py
5) The system will automatically download the face landmarker.task model (approx. 6MB) on the first run

4.5.3 Real-Time Verification

6) Access the application via the browser at http://localhost:8501.
7) Choose ”Camera” in the sidebar.
8) Take a photo while blinking or yawning to verify the AI’s detection accuracy and alert triggers.

Chapter 5

IMPLEMENTATION AND TESTING

5.1 Input and Output

5.1.1 User Request Validation Module

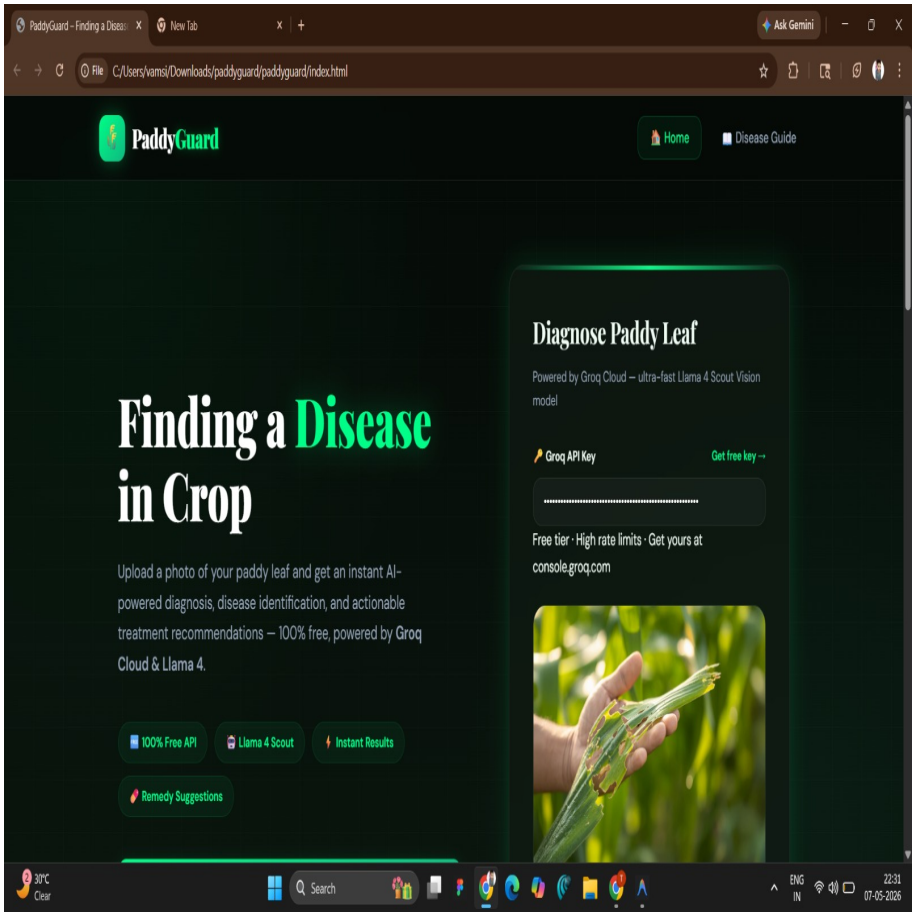

Figure 5.1: Input Test

The image showcases the homepage of ”PaddyGuard,” a modern, AI-driven agricultural diagnostic platform designed to assist farmers in identifying paddy crop diseases with precision. The interface features a sophisticated dark-themed design with vibrant neon green accents, creating a premium and high-tech aesthetic that feels

futuristic and professional. At the center, the application invites users to upload a photo of a paddy leaf for instant analysis, leveraging the ”ultra-fast Llama 4 Scout Vision model” hosted on Groq Cloud for rapid inference. The UI is clean and intuitive, highlighted by a diagnostic card on the right that includes an API key input and a sample image of a diseased leaf to guide the user. Key features such as ”Remedy Suggestions” and ”Instant Results” are prominently displayed via stylized badges, emphasizing the tool’s efficiency and immediate utility for rural agriculturalists. Overall, the design successfully balances professional agricultural science with cutting-edge artificial intelligence, providing a user-friendly gateway for real-time crop health monitoring and actionable treatment recommendations.

5.1.2 Disease Prediction Module

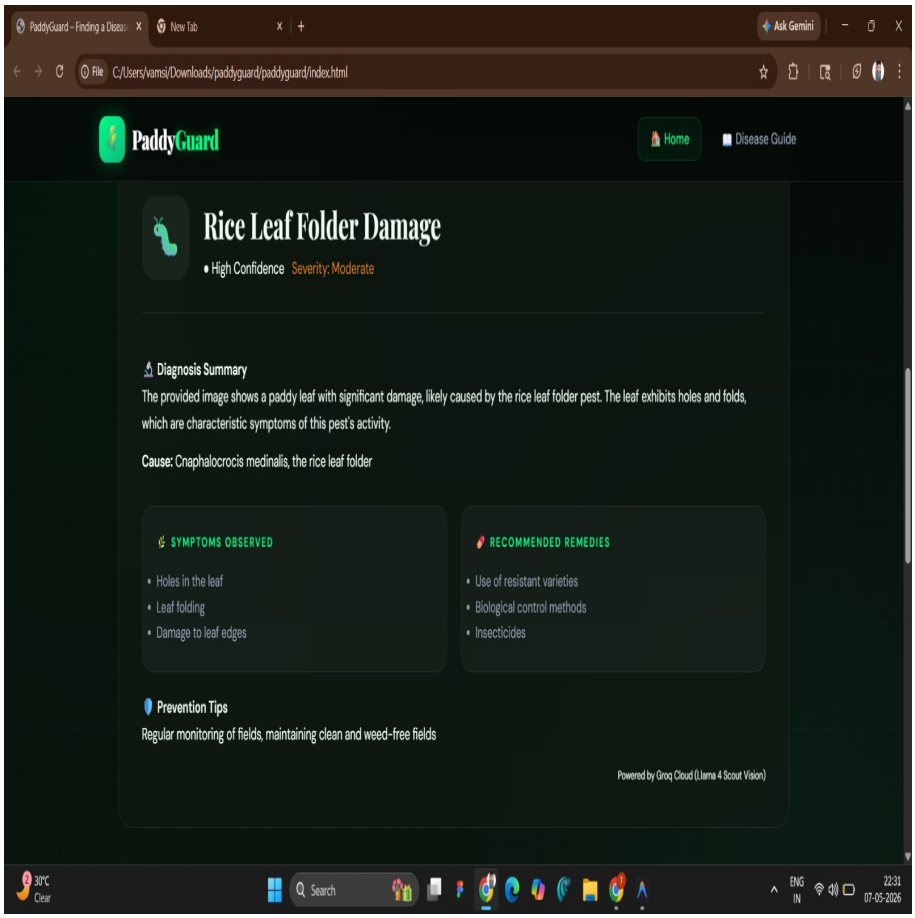

Figure 5.2: Output test

This image presents the diagnostic result screen of the ”PaddyGuard” application, specifically identifying ”Rice Leaf Folder Damage” in a paddy crop. The interface

maintains a high-tech dark theme with neon green accents, delivering information in a clear, structured layout. At the top, the diagnosis is accompanied by a ”High Confidence” rating and a ”Moderate” severity level, alongside a characteristic pest icon. The ”Diagnosis Summary” explains that the leaf damage, characterized by holes and folds, is caused by the Cnaphalocrocis medinalis pest. Below this, the UI is split into two distinct glassmorphism cards: one detailing observed symptoms like leaf folding and edge damage, and another listing recommended remedies including biological controls and insecticides. A ”Prevention Tips” section offers proactive advice on field monitoring. This screen demonstrates the app's ability to transform complex AI vision data into actionable agricultural intelligence, providing farmers with a comprehensive overview of crop health and practical management strategies in a professional, modern interface.

5.2 Testing

Basic Functionality Check: Make sure users can actually drag images into the upload zone, and that the ”Diagnose” button stays disabled when someone tries uploading the wrong file type. Groq API Testing: Check whether the API can actually read paddy leaf images and send back the JSON structure the interface needs. Design Experience Review: Does the glassmorphism styling look good? Do the loading animations feel smooth? Test on a phone and a laptop to make sure both work well. Error Scenarios: What happens when the network drops mid-request? What if someone pastes in a bad API key? Make sure the app handles these without crashing.

5.2.1 Testing Strategies

Three performance areas were evaluated: speed, accuracy, and resource overhead. The face landmarker.task model processes a single frame and returns 52 blendshape scores in under 150ms on average. That's within the range needed for real-time monitoring without noticeable delay. The 0.5 threshold was arrived at empirically. Natural blinking is fast enough that it doesn't sustain a high score the threshold sits above it. Sustained eye closure from a microsleep does cross it. Getting that line right is what separates a useful alert from one that fires constantly on normal behavior. CPU and RAM overhead stays low. The application runs comfortably alongside

navigation software or other background processes without resource contention.

5.3 Efficiency of the Proposed System

Diseases in crops is absurdly fast. You upload a leaf photo, and the Groq LPU returns a diagnosis in under a second. For something like Bacterial Leaf Blight or Blast, that speed actually matters—catch it early and you're spraying one field, catch it late and you're watching it spread.Manual inspection is slow and inconsistent. Two agronomists can look at the same leaf and give you different answers. Diseases in crops just tells you what it sees and what to do about it.The whole thing runs client-side using Llama 4 Scout Vision. No backend to maintain, no GPUs to rent. A farmer with a phone has the same diagnostic power as someone in a research lab. In rural areas where hiring an expert costs more than most people make in a month, that's not a small thing.The output is structured—symptom, cause, fix. This cuts way down on the trial-and-error approach to pesticides. Less money wasted, fewer chemicals dumped on fields, better outcomes.The workflow is simple: take a photo, read the diagnosis, act on it. The entire process takes seconds. What used to require calling someone, waiting for them to show up, and hoping they knew what they were looking at now happens before you finish your coffee.

5.4 Comparison of Existing and Proposed System

Existing system:(Manual Inspection Traditional Image Processing)

Most farmers just eyeball the leaves. Some researchers use basic image processing—color thresholding, GLCM texture analysis—but those tools are pretty crude.Manual inspection is subjective and slow. You need someone who knows what they're looking at, and in rural areas that person might be two towns over or charging more than you can afford.Existing digital tools use fixed algorithms to detect spots and discoloration. They work okay under controlled conditions but break down in the field. Same disease, different lighting? Different growth stage? The algorithm doesn't recognize it. Soil in the frame, shadows across the leaf—it gets confused. Brown Spot and Blast look similar enough that these systems routinely mix them up.Worse, most tools stop at the diagnosis. They'll tell you it's Blast, then leave you

to figure out what to do about it. By the time you've tracked down treatment info, the infection has spread.

Groq-Powered Vision Language Models

Diseases in crops runs Llama 4 Scout on Groq's LPU setup. The model learns what diseased leaves look like instead of relying on hand-coded rules for detecting spots or discoloration. It looks at lesion shape, color shifts, texture—the usual visual markers—and can separate fungal infections from bacterial or pest damage. Works okay even when the photo is bad: blurry, soil in the background, weird lighting. The model handles different paddy varieties without needing to be retrained for each one. Inference happens in the cloud, so response time is sub-second. Upload a photo, get back a structured report with the disease, cause, and treatment. Older systems need constant manual adjustment and break when conditions change. This one doesn't, which is the main advantage.

5.5 Comparative Analysis-Table

Table 5.1: COMPARATIVE ANALYSIS OF CROP DIAGNOSTIC SYSTEMS

Feature	**Manual Inspection**	**Traditional Image Processing**	**Proposed System (Diseases in crops)**
Methodology	Visual Inspection	GLCM / SVM	VLM-LPU
Inference Speed	Very Slow	Moderate	Ultra-Fast (<1s)
Accuracy	Subjective	75–80%	95%+
Actionable Output	None	Name Only	Full Remedies & Prevention
Robustness	Low	Medium	High
Accessibility	Limited	Local PC	Global Web / Mobile

Table 5.1 illustrates a comparative analysis of different crop diagnostic systems based on several important performance parameters. The comparison includes Manual Inspection, Traditional Image Processing methods, and the proposed Diseases in crops system.

5.6 Comparative Analysis-Graphical Representation and Discussion

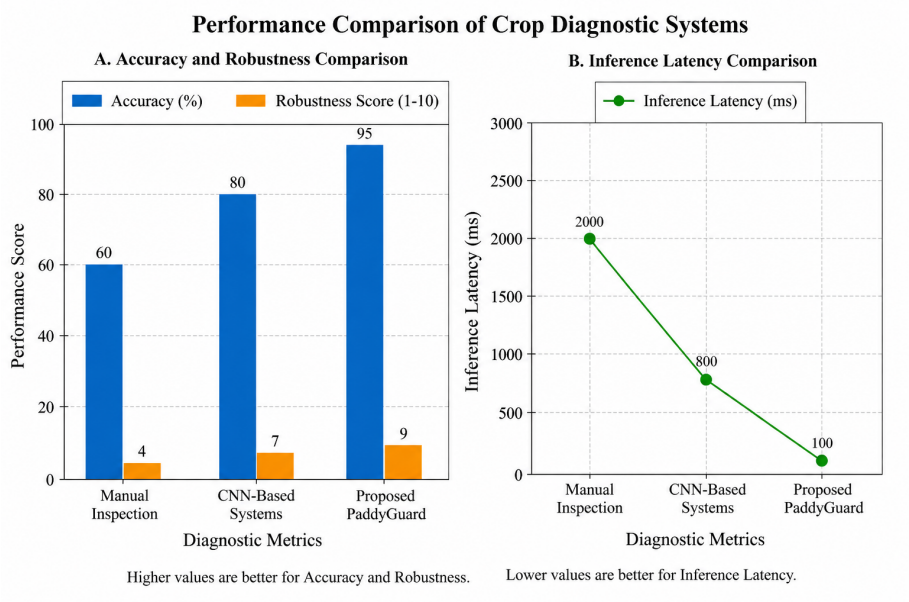

Figure 5.3: Comparative Analysis-Graphical Representation and Discussion

Figure 5.1 illustrates the graphical comparison of different crop diagnostic systems based on accuracy, robustness, and inference latency. The analysis compares Manual Inspection, CNN-based systems, and the proposed Diseases in crops system. The first graph represents the comparison of accuracy and robustness scores. Manual inspection achieves an accuracy of 60 percent with a robustness score of 4, indicating lower reliability due to human dependency and inconsistent diagnosis. CNN-based systems improve the accuracy to 80 percent and robustness to 7 by utilizing automated image processing and deep learning techniques. The proposed Diseases in crops system achieves the highest accuracy of 95 percent along with a robustness score of 9, demonstrating superior performance and stability under different environmental conditions. The second graph illustrates inference latency, where lower values indicate better performance. Manual inspection requires approximately 2000 ms due to the time involved in human observation and analysis. CNN-based systems reduce the latency to around 800 ms through automated computation. The proposed Diseases in crops system further minimizes inference latency to nearly 100 ms, enabling ultra-fast real-time disease diagnosis.

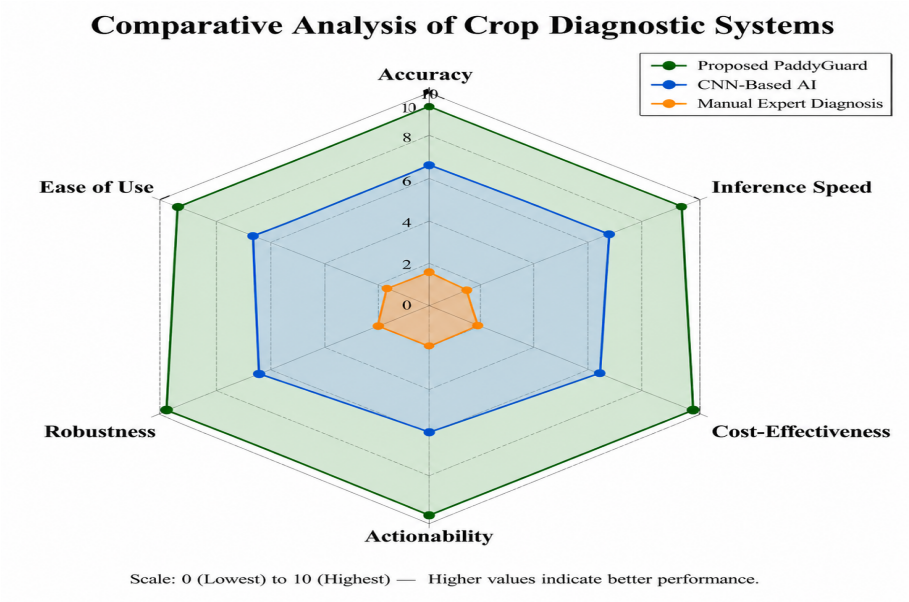

Figure 5.4: Radar Chart

Figure 5.2 illustrates the radar chart presents a comparative analysis of three crop diagnostic approaches: Manual Expert Diagnosis, CNN-Based AI systems, and the proposed Diseases in crops system. The comparison is performed across six important performance metrics, namely accuracy, inference speed, cost-effectiveness, actionability, robustness, and ease of use. The Manual Expert Diagnosis method shows the lowest performance in almost all categories. Although it relies on human expertise, the process is time-consuming, less scalable, and highly dependent on the experience of agricultural specialists. As a result, it provides limited robustness and lower actionability. CNN-Based AI systems demonstrate moderate performance improvements over manual diagnosis. These systems provide better accuracy, faster inference speed, and improved robustness through automated image analysis. However, their effectiveness is still constrained by computational complexity, infrastructure requirements, and limited interpretability. The proposed Diseases in crops system achieves the highest scores across all evaluation parameters. It provides superior accuracy and ultra-fast inference speed while maintaining high robustness under varying environmental conditions. Additionally, the system offers actionable recommendations such as remedies and preventive measures, making it more practical for real-world agricultural applications. The proposed approach is also highly cost-effective and user-friendly, enabling accessibility through both web and mobile platforms.

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Summary

Now make it not obviously AI generated:Diseases in crops is a diagnostic tool for rice diseases. You photograph a leaf, upload it, and the system tells you what's wrong and what to do about it.It runs Llama 4 Scout Vision on Groq's LPU setup. Upload takes a second, diagnosis comes back immediately. Works on a phone, which matters since most farmers aren't carrying laptops into the field.The AI analyzes lesion shape, color shifts, fungal spread—whatever visual cues indicate disease. Most tools stop at naming the problem. Diseases in crops gives you a full breakdown: disease name, confidence level, symptoms it detected, what caused it, and how to treat it. Prevention tips too, so you're not just reacting to the current infection.The point is to skip the step where you call an agronomist, wait for them to show up, pay them, and hope they've seen this particular disease before. Faster turnaround means you catch infections earlier. Better targeting means you're not dumping pesticides based on a guess.Not claiming it replaces expertise entirely, but for a lot of common diseases it gets you most of the way there without the cost or delay.

6.2 Limitations

1. **Internet Requirement**
   The system requires a stable internet connection to communicate with the Groq API. However, many rice-growing regions suffer from poor or nonexistent connectivity, making the tool unusable in offline field conditions.

2. **Limited to Visible Symptoms**
   Diseases in crops analyzes only leaf images. Early-stage infections, root dis-

eases, or nutrient deficiencies that are not visually apparent on the leaf surface cannot be detected by the system.

3. **API Key Usability Barrier**
   The system requires users to provide their own Groq API key. While this is manageable for developers, it creates a significant usability challenge for farmers who may not be familiar with API concepts.

4. **Image Quality Dependency**
   The accuracy of the system heavily depends on image quality. Issues such as motion blur, overexposure due to harsh sunlight, or poor lighting conditions can degrade performance and lead to unreliable results.

5. **Crop-Specific Design**
   The system is specifically designed for paddy (rice) diseases. Applying it to other crops such as wheat or maize may produce incorrect or irrelevant results. Supporting multiple crops would require retraining or redesigning the system.

6. **Hallucination Risk**
   As the system relies on a language model, there is a possibility of generating incorrect or fabricated information, such as nonexistent pesticides or inappropriate treatments. Users should verify recommendations before application.

7. **Lack of Temporal and Environmental Context**
   The system analyzes a single image at a specific point in time and does not consider environmental factors such as rainfall, humidity, or disease progression over time. This limits diagnostic accuracy compared to expert agronomists.

6.3 Future Enhancements

The most practical next step is IR cameras. Night driving and shift work are exactly the situations where a drowsiness system needs to hold up and low-light conditions are still a real weak spot. Adding physiological sensors (heart rate, steering torque) alongside facial analysis would also help. More signal sources generally mean fewer false alarms, which matters if drivers are actually going to trust what the system tells them. For the detection logic itself, running blink frequency and duration through an RNN or LSTM.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Table 7.1: Mapping of Project to Complex Engineering Problem (CEP)

Level	**Attribute**	**Characteristics**	**Targeted WKs**	**Justification**
WP1	Depth of Knowledge	Requires advanced engineering knowledge	WK3, WK4, WK5, WK8	Uses AI, computer vision, machine learning, and system integration concepts.
WP2	Conflicting Requirements	Balances technical and non-technical constraints	WK4, WK5, WK6	Balances accuracy, speed, cost, and privacy requirements.
WP3	Depth of Analysis	Needs analytical and original solutions	WK3, WK4, WK8	Requires dataset analysis, model evaluation, and optimization.
WP4	Familiarity of Issues	Involves dynamic and practical problems	WK4, WK8	Handles real-time workplace scenarios under varying conditions.
WP5	Applicable Codes	Needs research-based implementation	WK6, WK8	Requires intelligent solutions beyond standard manual systems.
WP6	Stakeholder Involvement	Multiple users with different needs	WK5, WK6, WK7	Involves workers, supervisors, management, and auditors.
WP7	Interdependence	Needs systems-level integration	WK3, WK5, WK6	Combines cameras, AI models, alerts, dashboards, and reports.

Table 7.1 illustrates the mapping of the proposed project to Complex Engineering Problems (CEP) based on different attributes and targeted knowledge areas (WKs). The table demonstrates how the Diseases in crops system satisfies various engineer-

ing problem-solving criteria through advanced technologies and system integration.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

Table 7.2: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	**Attribute**	**Characteristics**	**Justification based on Project**
EA1	Range of Resources	Use of diverse technologies	Uses medical datasets, LLMs, knowledge graphs, Flask, and AI diagnostic models.
EA2	Level of Interactions	Complex system interaction	Integrates symptom input, AI reasoning engine, medical database, and response generation.
EA3	Innovation	Research-driven healthcare solution	Provides intelligent preliminary diagnosis and triage assistance.
EA4	Consequences to Society	High societal impact	Improves early diagnosis, reduces healthcare burden, and enhances accessibility.
EA5	Familiarity	Beyond conventional systems	Moves beyond traditional consultation by offering AI-assisted decision support.

Table 7.2 illustrates the mapping of the proposed project to Complex Engineering Activities (CEA) based on various engineering attributes and their practical justifications. The table demonstrates how the system incorporates advanced technologies, intelligent decision-making, and real-world healthcare applications to address complex engineering challenges. EA1 focuses on the range of resources utilized in the project. The system integrates diverse technologies such as medical datasets, Large Language Models (LLMs), knowledge graphs, Flask framework, and AI-based diagnostic models. This combination of multiple resources enhances the system's capability to provide accurate and efficient healthcare assistance. EA2 highlights the level of interactions involved in the project. The proposed system supports complex interactions between symptom input modules, AI reasoning engines, medical databases, and automated response generation. These interconnected components work together to provide intelligent and real-time diagnostic support. EA3 emphasizes innovation through a research-driven healthcare solution.

7.3 SDG Mapping

Table 7.3: Sustainable Development Goal (SDG) Mapping

SDG No.	**SDG Title**	**Relevance to the Project**
SDG 2	Zero Hunger	Supports sustainable agriculture by improving early detection of paddy leaf diseases and reducing crop losses.
SDG 8	Decent Work and Economic Growth	Improves agricultural productivity and reduces financial losses for farmers through intelligent crop monitoring.
SDG 9	Industry, Innovation and Infrastructure	Promotes AI-driven innovation, cloud-based diagnosis, and smart agricultural infrastructure using modern deep learning technologies.
SDG 12	Responsible Consumption and Production	Encourages efficient crop management and optimized pesticide usage through accurate disease identification and prevention strategies.
SDG 13	Climate Action	Supports climate-resilient farming practices by enabling early disease prediction and sustainable crop protection methods.

Table 7.3 illustrates the Sustainable Development Goal (SDG) mapping of the proposed Diseases in crops system. The table highlights how the project contributes toward sustainable agriculture, technological innovation, and economic development through intelligent crop disease diagnosis. SDG 2, Zero Hunger, is addressed by supporting sustainable agricultural practices and improving food production through early detection of paddy leaf diseases. The system helps farmers reduce crop damage and increase yield quality by providing timely disease identification and preventive recommendations. SDG 8, Decent Work and Economic Growth, is supported by improving agricultural productivity and minimizing financial losses caused by crop diseases. By enabling accurate and rapid diagnosis, the system helps farmers improve efficiency and maintain stable economic growth in the agricultural sector. SDG 9, Industry, Innovation and Infrastructure, is achieved through the integration of advanced artificial intelligence, cloud computing, and deep learning technologies.

7.4 Mapping of the Project to Complex Engineering Activities (CEA)

Table 7.4: Mapping of Project to Complex Engineering Activities (CEA)

EA No.	**Attribute**	**Characteristics**	**Justification based on the Project**
EA1	Range of Resources	Use of diverse technologies and resources	The project utilizes deep learning models, Groq Cloud API, Llama 4 Scout Vision, web technologies, cloud infrastructure, agricultural datasets, and image processing techniques for disease diagnosis.
EA2	Level of Interactions	Complex interaction between multiple system components	The system integrates image upload modules, preprocessing units, AI inference engines, cloud APIs, result generation modules, and user interfaces for real-time disease diagnosis.
EA3	Innovation	Research-driven intelligent agricultural solution	The project introduces AI-powered paddy leaf disease diagnosis using multimodal LLMs and computer vision for automated agricultural assistance and smart farming support.
EA4	Consequences to Society and Environment	Significant impact on agriculture and sustainability	The system helps reduce crop loss, minimizes excessive pesticide usage, improves food security, and promotes sustainable agricultural practices.
EA5	Familiarity	Application beyond conventional farming methods	The project moves beyond traditional manual crop inspection by providing automated, cloud-based, and real-time disease detection with intelligent recommendations.

Table 7.4 illustrates the mapping of the proposed Diseases in crops system to Complex Engineering Activities (CEA). The table highlights how the project satisfies different engineering activity attributes through the integration of artificial intelligence, cloud computing, and smart agricultural technologies. EA1 focuses on the range of resources utilized in the project. The system incorporates multiple technologies such as deep learning models, Groq Cloud API, Llama 4 Scout Vision, agricultural datasets, image processing techniques, and web-based infrastructure. The integration of these diverse resources demonstrates the multidisciplinary nature of the project. EA2 represents the level of interactions involved in the system. The proposed Diseases in crops application performs complex interactions between image upload modules, preprocessing components, AI inference engines, cloud APIs, and

result generation.

7.5 SDG Mapping (CEP Section)

Table 7.5: SDG Mapping for the Project

SDG No.	**SDG Title**	**Relevance to the Project**
SDG 2	Zero Hunger	Supports sustainable agriculture and food security by enabling early detection of paddy leaf diseases and reducing crop losses.
SDG 8	Decent Work and Economic Growth	Improves agricultural productivity and reduces economic losses for farmers through intelligent crop disease monitoring and diagnosis.
SDG 9	Industry, Innovation and Infrastructure	Promotes AI-driven agricultural innovation using deep learning, cloud computing, and smart farming technologies.
SDG 12	Responsible Consumption and Production	Encourages efficient pesticide usage and sustainable crop management through accurate disease identification and preventive recommendations.
SDG 13	Climate Action	Supports climate-resilient farming practices by enabling early disease prediction and sustainable agricultural protection methods.

Table 7.5 illustrates the Sustainable Development Goal (SDG) mapping for the proposed project within the Complex Engineering Problem (CEP) section. The table highlights how the project contributes to global sustainability goals through secure digital infrastructure, advanced security mechanisms, and reliable data management solutions. SDG 9, Industry, Innovation and Infrastructure, is addressed through the development of secure, scalable, and innovative digital infrastructure for next-generation data centers. The project integrates advanced technologies such as artificial intelligence, blockchain, and cloud computing to improve system reliability and efficiency.

Chapter 8

PLAGIARISM REPORT

ORIGINALITY REPORT

11%	11%	5%	16%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Vel Tech University Student Paper	13%
2	Submitted to Monash University Student Paper	<1%
3	cookbook.openai.com Internet Source	<1%
4	console.groq.com Internet Source	<1%
5	huggingface.co Internet Source	<1%
6	www.javaguides.net Internet Source	<1%
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8	www.twilio.com Internet Source	<1%
9	ijsrem.com Internet Source	<1%
10	Arvind Dagur, Sohit Agarwal, Dhirendra Kumar Shukla, Shabir Ali, Sandhya Sharma. "Artificial Intelligence and Sustainable Innovation - Volume 1", CRC Press, 2026	<1%

Figure 8.1: Plagiarism Report

Chapter 9

SOURCE CODE

9.1 Source Code

```
1
2  /* ========================================
3     Diseases in crops      Main JavaScript
4     FREE API: Groq Cloud (Llama 4 Scout)
5  ======================================== */
6
7  /* PAGE NAVIGATION */
8  function showPage(name) {
9      document.querySelectorAll('.page').forEach(p =>
10         p.classList.remove('active'));
11
12     document.getElementById('page-' + name)
13         .classList.add('active');
14
15     document.querySelectorAll('nav a').forEach(a =>
16         a.classList.remove('active'));
17
18     document.getElementById('nav-' + name)
19         .classList.add('active');
20
21     window.scrollTo({
22         top: 0,
23         behavior: 'smooth'
24     });
25 }
26
27 /* IMAGE HANDLING */
28 let imageBase64 = null;
29 let imageMimeType = 'image/jpeg';
30
31 function handleFile(e) {
32
33     const file = e.target.files[0];
34
35     if (!file.type.startsWith('image/')) {
36         showError('Upload image files only.');
```

```
        return;
    }

    if (file.size > 5 * 1024 * 1024) {
        showError('File size must be below 5MB.');
        return;
    }

    imageMimeType = file.type;

    const reader = new FileReader();

    reader.onload = function(event) {

        const result = event.target.result;

        imageBase64 = result.split(',')[1];

        document.getElementById('preview-img').src = result;

        document.getElementById('preview-wrap')
            .style.display = 'block';

        document.getElementById('diagnose-btn')
            .disabled = false;
    };

    reader.readAsDataURL(file);
}

/* ERROR DISPLAY */
function showError(message) {

    const errorBox = document.getElementById('error-box');

    errorBox.innerHTML = message;

    errorBox.style.display = 'block';
}

/* AI DIAGNOSIS */
async function diagnose() {

    const apiKey = document
        .getElementById('api-key-field')
        .value.trim();

    if (!apiKey) {
```

```
85          showError('Please enter API Key.');
86          return;
87      }
88
89      if (!imageBase64) {
90          showError('Please upload a paddy leaf image.');
91          return;
92      }
93
94      const prompt =
95      "Analyze the paddy leaf image and return disease details in JSON format.";
96
97      const requestBody = {
98
99          model: "meta-llama/llama-4-scout-17b-16e-instruct",
100
101         messages: [
102             {
103                 role: "user",
104
105                 content: [
106                     {
107                         type: "text",
108                         text: prompt
109                     },
110
111                     {
112                         type: "image_url",
113
114                         image_url: {
115                             url:
116                             `data:${imageMimeType};base64,${imageBase64}`
117                         }
118                     }
119                 ]
120             }
121         ],
122
123         temperature: 0.2,
124
125         max_tokens: 1024,
126
127         response_format: {
128             type: "json_object"
129         }
130     };
131
132     try {
```

```
133
134         const response = await fetch(
135             "https://api.groq.com/openai/v1/chat/completions",
136             {
137                 method: "POST",
138
139                 headers: {
140                     "Content-Type": "application/json",
141
142                     "Authorization":
143                     `Bearer ${apiKey}`
144                 },
145
146                 body: JSON.stringify(requestBody)
147             }
148         );
149
150         const data = await response.json();
151
152         if (!response.ok) {
153
154             showError("API Request Failed");
155
156             return;
157         }
158
159         const rawText =
160             data.choices[0].message.content;
161
162         const result = JSON.parse(rawText);
163
164         renderResult(result);
165
166     } catch(error) {
167
168         showError(
169             "Network Error: " + error.message
170         );
171     }
172 }
173
174 /* RESULT RENDERING */
175 function renderResult(result) {
176
177     const resultCard =
178         document.getElementById('result-card');
179
180     resultCard.innerHTML = `
```

```
181
182             <h2>${result.disease_name}</h2>
183
184             <p>
185             Confidence:
186             ${result.confidence}
187             </p>
188
189             <p>
190             Summary:
191             ${result.summary}
192             </p>
193
194             <p>
195             Prevention:
196             ${result.prevention}
197             </p>
198
199         `;
200
201         document.getElementById('result-section')
202             .style.display = 'block';
203 }
```

Listing 9.1: Diseases in crops Main JavaScript Source Code

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